SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-033: Last Login Tracking and Display for Student Accounts

Field	Details
Change Request ID	CR033
Change Request Title	Last Login Tracking and Display for Student Accounts
Student Name	AAM. Nusky
Student ID	JF/ICT/24/01
Implementation Status	Implementation report prepared; local verification required
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

The administrator Students table does not show when a student last logged in. This change records the date and time of a successful student login and displays it in the administrator Students table, helping administrators identify recent account activity and students who have never logged in.

1.2 Existing Behaviour

- The login process validates the username and password and creates a JWT token after successful authentication.
- The users table stores account and role information, but the Students list does not display a last-login timestamp.
- The administrator Students table displays Registration No., Name, Email, Phone, and Actions.
- Administrators cannot currently see the most recent successful login date and time.

1.3 Requested Change

- Add a nullable last_login DATETIME field to the users table.
- Update last_login only after username and password verification succeeds.
- Store the current database date and time using NOW().
- Return last_login with student records through the existing student API.
- Display a Last Login column in the administrator Students table.
- Format the timestamp into a readable date and time in the frontend.
- Display “Never logged in” when no login timestamp exists.
- Preserve existing student Add, Edit, Delete, authentication, and authorization functionality.

2. Implementation

2.1 Implementation Summary

The CR033 change covers the database, backend authentication flow, student data retrieval, and administrator frontend. A nullable last_login column is added to users. After bcrypt password verification succeeds, updateLastLogin() stores the timestamp using NOW(). Student queries join students with users so last_login is returned to the frontend. DashboardPage.jsx displays a Last Login column and shows “Never logged in” when the value is null.

2.2 Affected Files

File	Change / Purpose
database/schema.sql	Add nullable last_login DATETIME column to users for new schema creation.
database/migrations/CR033_add_last_login.sql	Migration for an existing database; run once only if last_login is absent.
backend/src/models/userModel.js	Add updateLastLogin() and export it with existing user model functions.
backend/src/controllers/authController.js	Call updateLastLogin() after successful bcrypt password verification.
backend/src/models/studentModel.js	Include users.last_login in student queries using a JOIN.
frontend/src/pages/DashboardPage.jsx	Add Last Login heading and render readable date/time or “Never logged in”.

2.3 Technical Changes

Database Field

The users table stores last_login as DATETIME NULL DEFAULT NULL. NULL represents an account with no recorded successful login.

Successful Login Timestamp

The user model runs an UPDATE query with SET last_login = NOW() and a parameterized user_id. The update is invoked only after credentials are verified.

Student API Data

Student retrieval queries select users.last_login by joining students.user_id to users.user_id. Password hashes and other authentication secrets must not be exposed in student responses.

Frontend Display

The Students table includes a Last Login column. A recorded timestamp is formatted with JavaScript toLocaleString(); a null or missing value is displayed as “Never logged in”.

Existing Flow Preservation

- Password verification continues to use bcrypt.
- The existing JWT authentication and authorization flow should remain unchanged.
- Existing student Add, Edit, and Delete actions should continue to use student_id.
- No login history, password reset, new authentication method, or JWT expiration change is included.

2.4 Implementation Logic

1. Add the nullable last_login field to users.
2. Receive username and password in the existing login handler.
3. Verify the password with bcrypt; reject invalid credentials without updating last_login.
4. After successful verification, call updateLastLogin(user.user_id), which stores NOW().
5. Continue the existing token and login-response flow.
6. Return last_login in student records through the existing student API.
7. Render the Last Login column in the administrator Students table.
8. Display “Never logged in” when no timestamp is recorded.

2.5 Error and Data Handling

- Invalid credentials must not update last_login.
- The database update uses a parameterized user ID.
- The student API must not expose password hashes or other sensitive authentication data.
- A null last_login value is handled by the frontend fallback text.
- Run the migration only when the column does not already exist.

3. Testing & Verification

3.1 Testing Approach

Recommended verification in the local development environment includes functional, positive and negative login, API response, UI, and CRUD regression testing. The cases below are acceptance checks to execute; they are not represented as passed without local evidence.

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Inspect users schema	last_login exists as nullable DATETIME.	Pending local verification	Not verified
TC-02	Successful student login	last_login updates to current database time.	Pending local verification	Not verified
TC-03	Incorrect password	Login rejected; last_login unchanged.	Pending local verification	Not verified
TC-04	Load student list API	Student records include last_login.	Pending local verification	Not verified
TC-05	Student with NULL timestamp	Dashboard displays “Never logged in”.	Pending local verification	Not verified

TC-06	Student with timestamp	Dashboard displays readable date and time.	Pending local verification	Not verified
TC-07	Existing Edit action	Correct student can still be edited.	Pending local verification	Not verified
TC-08	Existing Delete action	Correct student can still be deleted.	Pending local verification	Not verified
TC-09	Authentication / authorization	Existing JWT and administrator rules remain effective.	Pending local verification	Not verified

3.2 Test Summary

Test Result Count	Result
Total test cases	9
Passed	To be completed after local testing
Failed	To be completed after local testing
Pass rate	Not yet calculated
Overall test result	Pending local verification

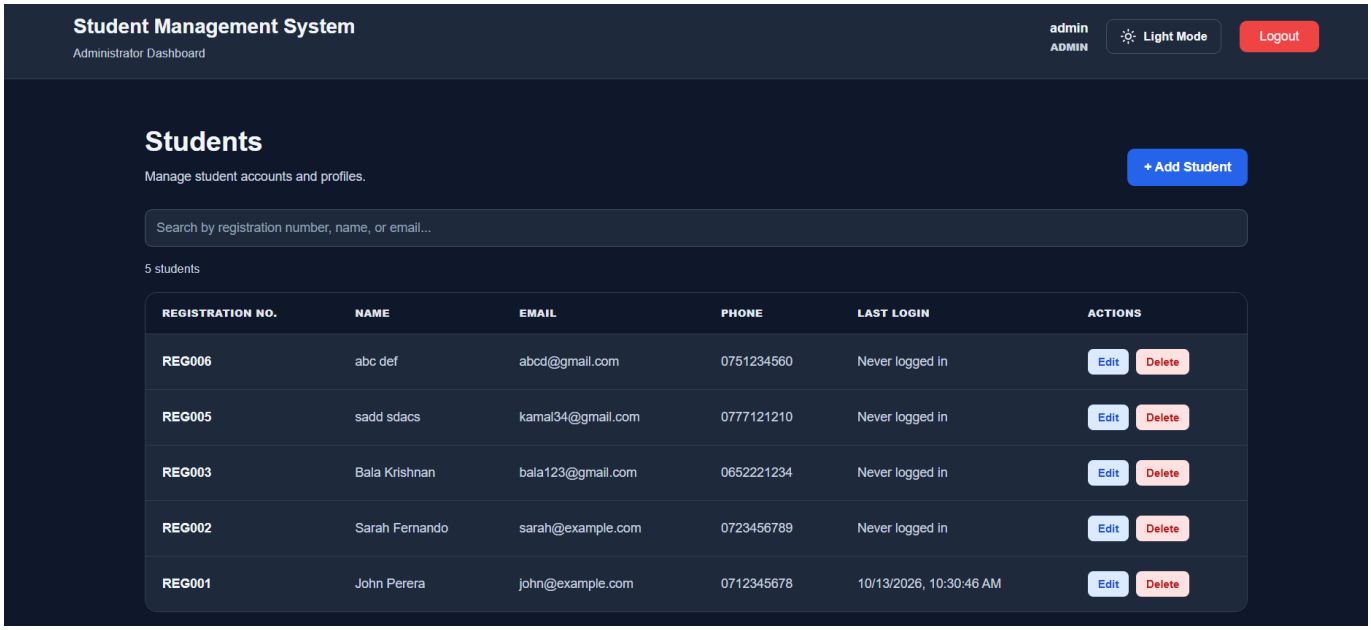
4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	Successful login records latest login date/time.	Check users.last_login after valid login.	Pending
AC-002	Invalid credentials do not update last_login.	Compare timestamp before and after failed login.	Pending
AC-003	Students table displays Last Login.	Inspect administrator dashboard.	Pending
AC-004	No-login students display “Never logged in”.	Check a student with NULL last_login.	Pending
AC-005	Existing Edit/Delete operations continue to work.	Perform CRUD regression tests.	Pending
AC-006	Authentication and authorization remain unchanged.	Test existing roles and JWT flow.	Pending

5. Evidence

Attach up to four screenshots after running the implementation and tests:

- Evidence 01 – Administrator Students table showing the Last Login column.
- Evidence 02 – Successful student login followed by the updated last_login value.
- Evidence 03 – Student with no recorded login showing “Never logged in”.
- Evidence 04 – Existing Edit/Delete actions working after the change.



6. Final Result

Item	Result
Requirements	Implementation scope defined according to CR033.
Implementation	Code changes described; local execution must be verified.
Testing	Pending local verification.
Acceptance Criteria	Pending verification against the running application.
Existing Login Flow	Designed to be preserved; regression test required.
Existing Student CRUD	Designed to be preserved; regression test required.
Final Status	Pending local verification

6.1 Final Summary

CR033 adds visibility of the most recent successful student login to the administrator Students table. The change adds a nullable database timestamp, updates it only after successful password verification, returns it with student records, and displays a readable date/time or “Never logged in”. The intended scope preserves the existing login, JWT, authorization, and student CRUD flows. Final completion and test results should be recorded after applying the changes and verifying them locally.

6.2 Issues / Limitations

This report does not claim tests have passed without local evidence. Confirm the actual database name, table columns, model exports, and login controller structure before applying code. If the database already exists, run the migration only if last_login is absent.

Student Name: AAM. Nusky

Student ID: JF/ICT/24/01

Date: 13october2026

Signature: _____

~ END OF REPORT ~